Princess Sumaya University for Technology - King Hussein School for Computing Sciences

CS11313 - **Algorithms Design and Analysis** - **Fall** 2021

Quiz # 1

Name:	**ID:**

Q1. Assume the running time of some algorithm satisfies the following relations: $T(n) = \omega(n^2\sqrt{n})$ and $T(n) = O(n^3 \log n)$, which of the following functions can $T(n)$ possibly be? Circle all that apply. n^2 *$n^{2.7}$ *n^3 $n^{3.1}$ $n^2\sqrt{n}$ $(\sqrt{n})^7$ **[2 points]**

Answer. The chosen functions must have an order of growth that is $\leq n^3 \log(n)$ and $> n^2\sqrt{n}$.

Q2. Fill in the following table. Each box should have a "Yes" or "NO" indicating whether or not the condition at the top of the row is true. **[3 points]**

A	B	$A = O(B)$	$A = \Omega(B)$	$A = \Theta(B)$
$\log(n!)$	$n\sqrt{n}$	YES	NO	NO
$\log(n)$	$\log(\sqrt{n})$	YES	YES	YES

Answer. Note that $\log(n!) = \Theta(n \log n)$ and $\log(\sqrt{n}) = \frac{1}{2}\log(n)$

Q3. Rearrange the following functions based on their order of growth (slowest growth to fastest growth). 2^n $\log_2(4^n)$ $4^{\log_2(n)}$ $\log(n!)$ **[2 points]**

Answer. 4 1 3 2

Note that $\log_2(4^n) = n\log_2(4) = 2n$ and $4^{\log_2(n)} = n^{\log_2(4)} = n^2$.

Q4. The definition of $f = O(g)$ covered in class used two constants named c and n_0. What is an example of a value for c and a value for n_0 that can be used to show that $3n + \log_2(n) = O(n)$?

Answer. c = 4 and n_o = 1 (other values are also acceptable) **[2 points]**

Q5. Which of the following is guaranteed to be true if $f_1 = O(g)$ and $f_2 = O(g)$? **[1 point]**

A $f_1 + f_2 = O(g)$ B. $f_1 = O(f_2)$ C. $f_1 + f_2 = \Omega(g)$ D. None of the answers